

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development using MERN
PRACTICAL - VI

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Class: TYIT	Batch: 2
Date of Assignment: 09-09-26	Date/Time of Submission: 09-09-26

6. Node.js and Express Fundamentals

a. Install Express and create a simple server.

Step 1 - create folder Expresspractical and open that folder in V S Code

Step 2 - in V S code open terminal -> (write) `npm init -y` => this will create package.json file in folder.

Step 3 - `npm install express` => this will install express

Step 4 - create server.js page in Expresspractical folder like this

```
ExpressPractical
|
├─ node_modules
├─ package.json
├─ package-lock.json
└─ server.js
```

Step 5 - Write following code into server.js file

server.js :

```
const express = require("express");
const app = express();
const PORT = 3000;
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});
```

Step 6 - run the code => open terminal & write => `node server.js`

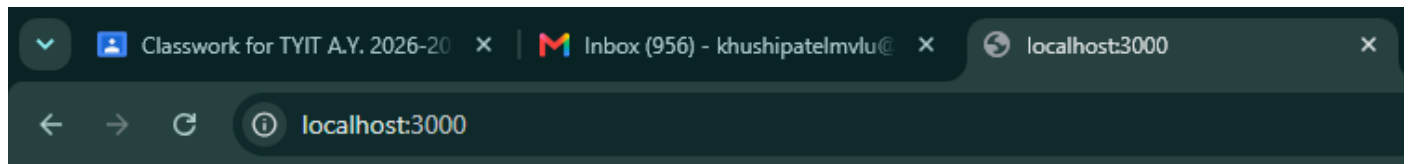
Step 7 - output in terminal => Server is running on <http://localhost:3000>

```
Node.js v24.17.0
PS C:\Users\info\OneDrive\Desktop\Mern Stack Practical\Expresspractical> node server.js
Server is running on http://localhost:3000
█
```

Step 8 - open browser & type => <http://localhost:3000>

b. Display a message on the browser using the server response.**server.js :**

```
import express from 'express';
const app = express();
const PORT = 3000;
app.get('/', (req, res) => {
  res.send('Hello from Express Server!.....KHUSHI ');
});
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});
```

OUTPUT:-**c. Create different routes in Express.****CODE:-**

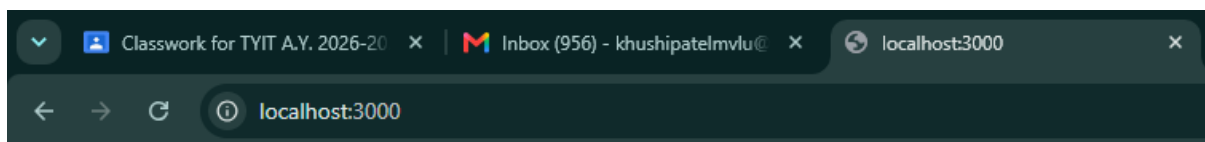
```
import express from 'express';
const app = express();
const PORT = 3000;
// Home Route
app.get("/", (req, res) => {
  res.send(`
<h1>MVLU College Management Server</h1>
<p>Welcome to our college website.</p>
`);
});
// Students Route
app.get("/students", (req, res) => {
  res.send(`
<h1>Student List</h1>
<ul>
<li>Rahul Sharma</li>
<li>Priya Patil</li>
<li>Amit Joshi</li>
<li>Neha Bhosale</li>
</ul>
`);
});
// Faculty Route
app.get("/faculty", (req, res) => {
  res.send(`
<h1>Faculty List</h1>
<ul>
<li>Sneha Gokarnkar - Information Technology</li>
`);
});
```

```

<li>Rohini Bhosale - Information Technology</li>
<li>Sumit Tripathi - Information Technology</li>
<li>Chetana Tanavade - Information Technology</li>
</ul>
`);
});
// Courses Route
app.get("/courses", (req, res) => {
res.send(`
<h1>Available Courses</h1>
<ul>
<li>B.Sc. Information Technology</li>
<li>B.Sc. Computer Science</li>
<li>B.Sc. Biotechnology</li>
<li>B.A. Multimedia and Mass Communication</li>
</ul>
`);
});
// Contact Route
app.get("/contact", (req, res) => {
res.send(`
<h1>Contact Us</h1>
<p>Email: info@mvlucollege.in</p>
<p>Phone: 9876543210</p>
<p>Address: Andheri, Mumbai</p>
`);
});
// Start Server
app.listen(PORT, () => {
console.log(`Server is running on http://localhost:${PORT}`);
});

```

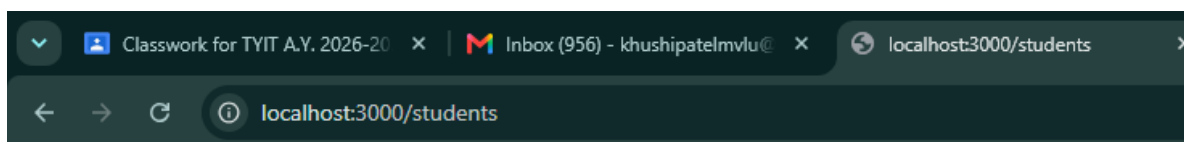
OUTPUT:-



MVLU College Management Server

Welcome to our college website.

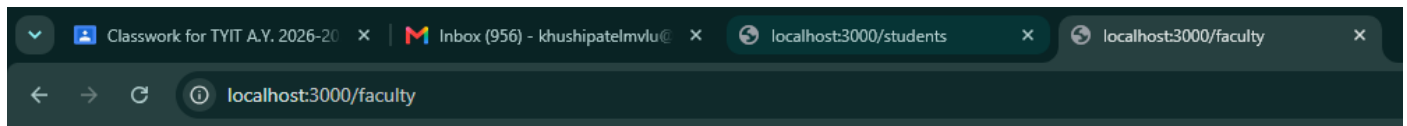
<http://localhost:3000/students>



Student List

- Rahul Sharma
- Priya Patil
- Amit Joshi
- Neha Bhosale

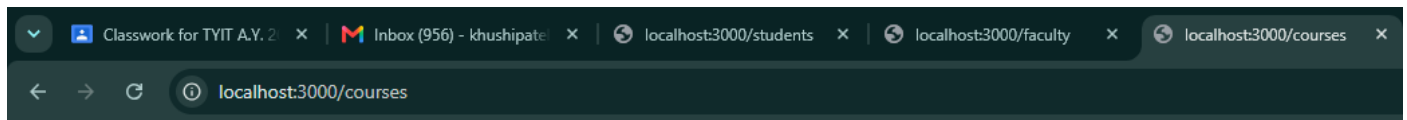
<http://localhost:3000/faculty>



Faculty List

- Sneha Gokarnkar - Information Technology
- Rohini Bhosale - Information Technology
- Sumit Tripathi - Information Technology
- Chetana Tanavade - Information Technology

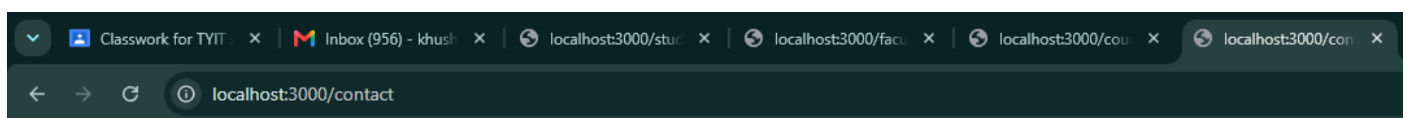
<http://localhost:3000/courses>



Available Courses

- B.Sc. Information Technology
- B.Sc. Computer Science
- B.Sc. Biotechnology
- B.A. Multimedia and Mass Communication

<http://localhost:3000/contact>



Contact Us

Email: info@mvlucollege.in

Phone: 9876543210

Address: Andheri, Mumbai

d. Send a JSON response from the server.

CODE:-

```
const express = require("express");
const app = express();
const PORT = 3000;
app.get("/", (req, res) => {
  res.send("Welcome to MVLU College Management Server");
});
app.get("/students", (req, res) => {
  const students = [
    {
      id: 1,
      name: "Rahul Sharma",
      course: "B.Sc. IT",
      marks: 85
    },
    {
      id: 2,
      name: "Priya Patil",
```

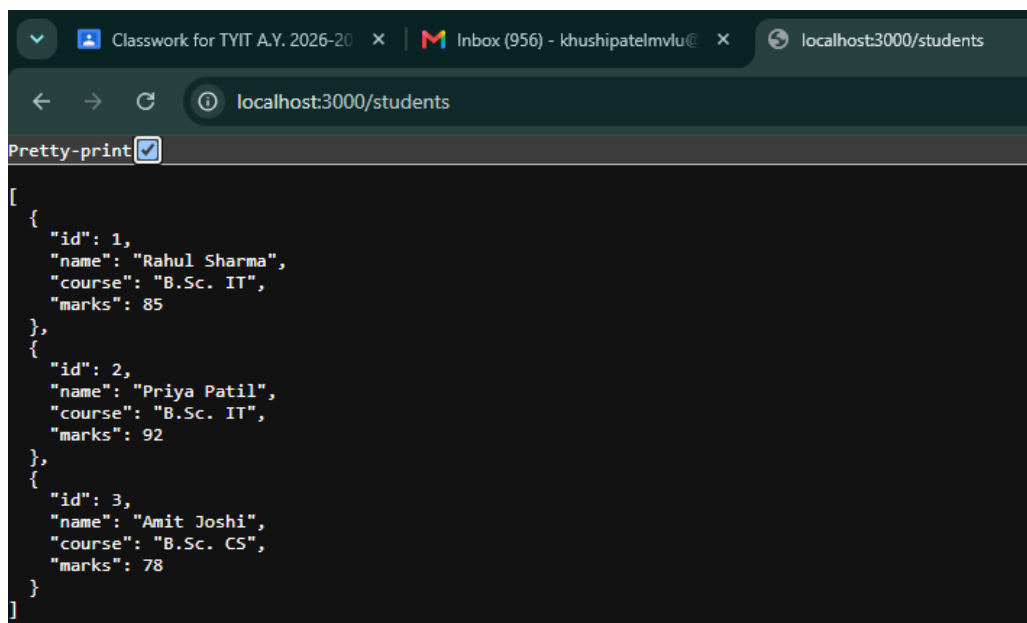
```

course: "B.Sc. IT",
marks: 92
},
{
id: 3,
name: "Amit Joshi",
course: "B.Sc. CS",
marks: 78
}
];
res.json(students);
});
app.listen(PORT, () => {
console.log(`Server is running on http://localhost:${PORT}`);

});

```

OUTPUT:-



e. Test routes using browser or Postman.

CODE:-

```

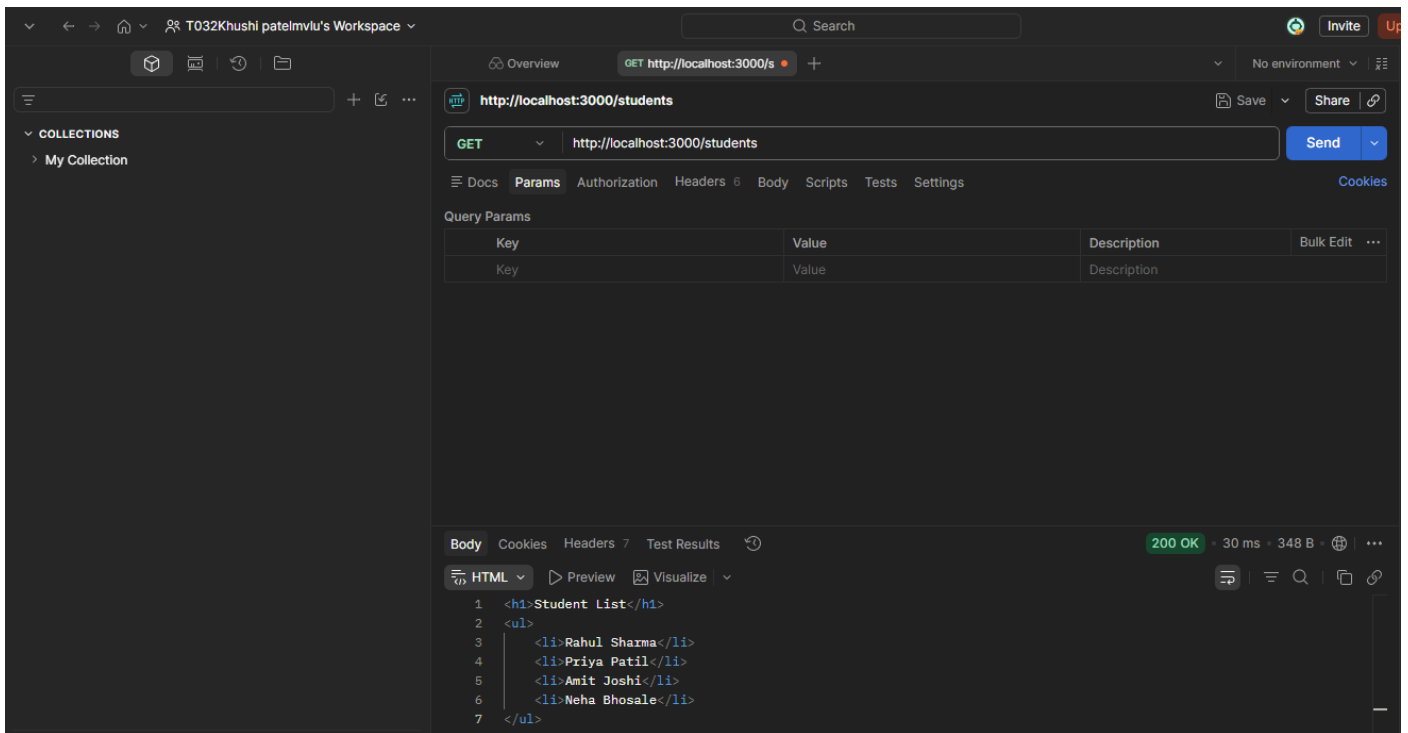
const express = require("express");
const app = express();
const PORT = 3000;
// Home Route
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res.send(`
<h1>MVLU College Management Server</h1>
<p>Welcome to our college website.</p>
`);
});
// Students Route
app.get("/students", (req, res) => {
res.send(`
<h1>Student List</h1>
<ul>

```

```

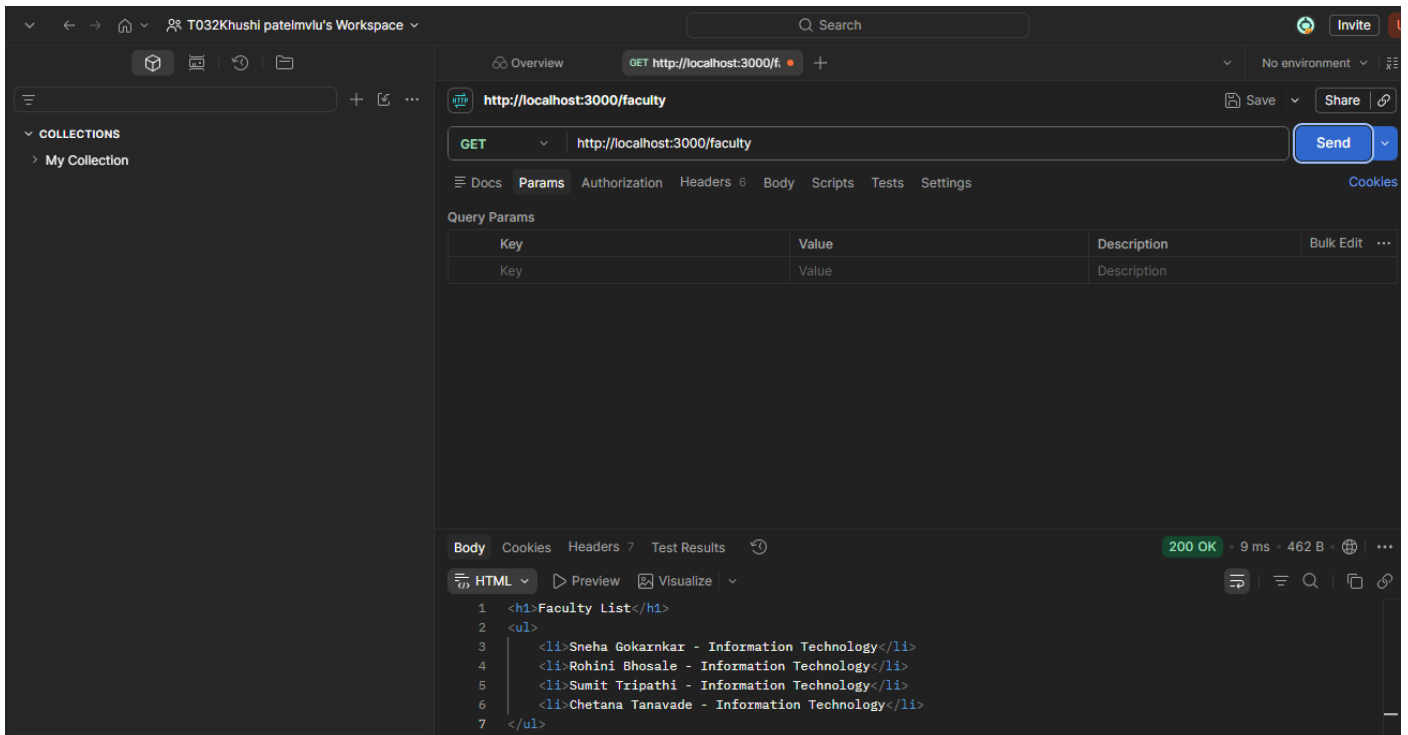
<li>Rahul Sharma</li>
<li>Priya Patil</li>
<li>Amit Joshi</li>
<li>Neha Bhosale</li>
</ul>
`);
});
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<h1>Available Courses</h1>
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<li>B.Sc. Biotechnology</li>
<li>B.A. Multimedia and Mass Communication</li>
</ul>
`);
});
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<h1>Contact Us</h1>
<p>Email: info@mvlucollege.in</p>
<p>Phone: 9876543210</p>
<p>Address: Andheri, Mumbai</p>
`);
});
// Start Server
app.listen(PORT, () => {
console.log(`Server is running on http://localhost:${PORT}`);
});
OUTPUT:-

```



Postman interface showing a GET request to `http://localhost:3000/students`. The response is 200 OK, 30 ms, 348 B. The response body is HTML:

```
1 <h1>Student List</h1>
2 <ul>
3   <li>Rahul Sharma</li>
4   <li>Priya Patil</li>
5   <li>Amit Joshi</li>
6   <li>Neha Bhosale</li>
7 </ul>
```



Postman interface showing a GET request to `http://localhost:3000/faculty`. The response is 200 OK, 9 ms, 462 B. The response body is HTML:

```
1 <h1>Faculty List</h1>
2 <ul>
3   <li>Sneha Gokarnkar - Information Technology</li>
4   <li>Rohini Bhosale - Information Technology</li>
5   <li>Sumit Tripathi - Information Technology</li>
6   <li>Chetana Tanavade - Information Technology</li>
7 </ul>
```

Overview GET http://localhost:3000/C +

http://localhost:3000/Contact

GET http://localhost:3000/Contact

Params Authorization Headers 6 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body Cookies Headers 7 Test Results

200 OK • 9 ms • 341 B •

HTML Preview Visualize

```

1 <h1>Contact Us</h1>
2 <p>Email: info@mvlucollege.in</p>
3 <p>Phone: 9876543210</p>
4 <p>Address: Andheri, Mumbai</p>

```

Overview GET http://localhost:3000/C +

http://localhost:3000/Courses

GET http://localhost:3000/Courses

Params Authorization Headers 6 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body Cookies Headers 7 Test Results

200 OK • 71 ms • 415 B •

HTML Preview Visualize

```

1 <h1>Available Courses</h1>
2 <ul>
3   <li>B.Sc. Information Technology</li>
4   <li>B.Sc. Computer Science</li>
5   <li>B.Sc. Biotechnology</li>
6   <li>B.A. Multimedia and Mass Communication</li>
7 </ul>

```

f. Run the server and observe the output in the console.

```

Server is running on http://localhost:3000
❖ PS C:\Users\info\OneDrive\Desktop\Wern Stack Practical\ep6> node --watch server.js
Server is running on http://localhost:3000

```